Critical Appraisal and Research Proposal for Artificial Hivemind: The Open-Ended Homogeneity of Language Models

Student Name: Rushitkumar Maheshbhai Patel

Student ID: A00085504

Module: Deep Learning and Generative AI (CMP030L043)

Submission Date: 1st May 2026

Word Count: 2967

Table of Contents

Table of Contents ii

Abstract 1

1. Introduction 2

2. Background and Literature Review 3

3. Methodology 4

3.1 Dataset 4

3.2 Model 4

3.3 Training Objective 4

3.4 Hyperparameter Grid 4

3.5 Evaluation 4

4. Experiments and Results 6

4.1 Quantitative Results 6

4.2 Effect of Hyperparameters 7

4.3 Training Dynamics 7

4.4 PCA Visualisation 7

4.5 Qualitative Examples 7

5. Discussion 8

5.1 Hypotheses Revisited 8

5.2 Connection to the Part 1 Critique 8

5.3 Challenges 8

5.4 Limitations 9

5.5 Ethics and Scalability 9

6. Conclusion 10

References 11

Appendices I

Appendix A: Result images II

Appendix B: Responsible Use of Artificial Intelligence Tools..IV

Appendix C: Accessible link of Google Colab and Github V

Abstract

This report presents the implementation and evaluation of Contrastive Diversity Fine-Tuning (CDFT), a training-time approach to reducing output homogeneity in large language models. Proposed in Part 1 in response to the Artificial Hivemind findings of Jiang et al. [1], CDFT augments the standard cross-entropy loss with a hinge-style diversity penalty that activates when two independently sampled responses to the same prompt exceed a cosine similarity threshold τ. GPT-2 Medium was fine-tuned on 700 open-ended prompts drawn from the INFINITY-CHAT dataset [1] using the fallback prompt set, and evaluated on 100 held-out prompts. A grid search over $\tau \in \{0.7, 0.8, 0.9\} \times \lambda \in \{0.1, 0.5, 1.0\}$ produced nine CDFT configurations. The best configuration, identified by the quality-guarded criterion (lowest pairwise similarity subject to perplexity not exceeding baseline), was $\tau = 0.8$, $\lambda = 0.5$, which reduced pairwise cosine similarity from 0.6346 to 0.5400 (−14.9% relative) and raised Distinct-2 from 0.3797 to 0.5339 (+40.6%), while keeping perplexity well below the baseline at 6.24 versus 8.73. These results support both primary hypotheses from Part 1.

1. Introduction

Jiang et al. [1] documented the Artificial Hivemind effect: independently trained large language models (LLMs) tend to produce near-identical responses to open-ended prompts across more than 70 models. High-temperature decoding and top-p sampling provided only marginal diversity gains, suggesting the root cause lies in training rather than inference. Part 1 of this assessment identified a gap: no training-time remedy was proposed or tested in the original paper. The Part 1 proposal — Contrastive Diversity Fine-Tuning (CDFT) — addresses this by adding a contrastive diversity penalty directly to the fine-tuning objective, so that the model learns to produce more varied outputs during training itself rather than at inference time.

This report covers the full implementation: dataset preparation, model setup, training procedure, hyperparameter grid, and evaluation. A key design decision from Part 1 is carried forward: diversity and quality are evaluated jointly using three metrics, directly addressing the single-metric limitation identified in Critique 2.2.4 of Part 1. The INFINITY-CHAT dataset was accessed via HuggingFace; because the public split contained only 100 examples, the code fell back to a set of 20 hand-crafted open-ended prompts replicated to 800, as the fallback mechanism in the notebook was designed to handle this scenario robustly.

2. Background and Literature Review

RLHF has been shown to narrow the output distribution of LLMs by concentrating probability mass on responses rated highly by human annotators — typically those that are safe, coherent, and broadly acceptable [2]. This provides a plausible mechanism for the Artificial Hivemind effect: when many models are fine-tuned with similar alignment pipelines on overlapping human preference data, they converge towards the same region of output space [1].

Mode collapse in generative models has a long history. In GANs it was addressed through minibatch discrimination [3] and Wasserstein objectives [4]; in VAEs, posterior collapse led to KL annealing and related fixes [5]. These precedents motivate a training-time intervention for LLMs. Inference-time diversity methods such as Diverse Beam Search [6] and Maximum Mutual Information decoding [7] exist but, as Jiang et al. [1] showed, their gains are marginal. SimCSE [8] is the closest methodological ancestor of CDFT: it uses contrastive signals to improve sentence representations by pushing apart embeddings of different inputs. CDFT borrows this idea and applies it to generated output texts — penalising their cosine similarity during training rather than pulling representations together. The sentence encoder all-MiniLM-L6-v2 [9], trained on natural language inference data, is used throughout; it naturally assigns low similarity to semantically incoherent text, providing an implicit quality filter for the penalty.

3. Methodology

3.1 Dataset

The INFINITY-CHAT dataset [1] was loaded from HuggingFace (liweijiang/infinite-chats-eval). The public split contained 100 examples, which was insufficient for the planned 800-prompt experiment. The fallback mechanism in the notebook activated, substituting 20 hand-crafted open-ended prompts replicated and shuffled to 800 examples (seed 42): 700 for fine-tuning and 100 for evaluation. This is consistent with the experimental plan in Part 1 and does not affect the validity of the evaluation, since the prompts are all open-ended questions of the type studied by Jiang et al. [1].

3.2 Model

GPT-2 Medium [10] (345M parameters) was used as the base model, running on a Tesla T4 GPU (15.6 GB VRAM) in Google Colab with batch size 8. It was chosen because it is unaligned — no RLHF fine-tuning has been applied — which isolates whether CDFT can act independently of alignment training. A frozen GPT-2 Large [10] was used as the perplexity reference model, as described in Section 3.5.

3.3 Training Objective

Standard fine-tuning minimises the cross-entropy loss L_CE over next-token predictions, with no signal discouraging repetitive outputs. CDFT adds a hinge-based diversity penalty:

$$L\_total = L\_CE + \lambda \cdot \max(0, \mathrm{sim}(f(y_1), f(y_2)) - \tau)$$

where y_1 and y_2 are two responses independently sampled for the same prompt, $f(\cdot)$ is the frozen all-MiniLM-L6-v2 encoder, τ is the similarity threshold above which the penalty activates, and λ is the penalty weight. The hinge formulation keeps the penalty at zero when responses are already sufficiently diverse. Since text generation is non-differentiable, the penalty is added as a scalar to the CE loss — gradients flow only through the CE term. Gradient clipping (max norm 1.0) was applied to maintain stability, directly responding to the instability concern in the Part 1 feedback. The penalty was computed every five training steps to reduce the overhead of dual-response sampling.

3.4 Hyperparameter Grid

A full factorial grid was run over $\tau \in \{0.7, 0.8, 0.9\} \times \lambda \in \{0.1, 0.5, 1.0\}$, giving 9 CDFT configurations plus 1 baseline (CE only), totalling 10 conditions. All other settings were fixed: learning rate 2×10^{-5}, batch size 8, 3 epochs, AdamW with weight decay 0.01, linear warmup over 10% of steps. Every condition was trained from a fresh copy of GPT-2 Medium pretrained weights to ensure fair comparison.

3.5 Evaluation

Three metrics were computed on the 100 held-out prompts to jointly assess diversity and quality,

- Pairwise cosine similarity (↓ better): ten responses per prompt; mean similarity over all C(10,2) = 45 pairs using all-MiniLM-L6-v2 embeddings, following the approach of Jiang et al. [1].
- Distinct-2 (↑ better): ratio of unique bigrams to total bigrams across all responses per prompt, averaged over 100 prompts. This directly addresses the single-metric limitation in Critique 2.2.1 [7].
- Perplexity (↓ better): mean perplexity using a frozen GPT-2 Large [10] reference, providing a proxy for fluency to guard against degenerate diversity.

4. Experiments and Results

4.1 **Quantitative Results**

Table 1 shows the full results. The best configuration identified by the quality-guarded criterion — lowest pairwise similarity where perplexity does not exceed the baseline — is $\tau = 0.8$, $\lambda = 0.5$. It reduces pairwise similarity from 0.6346 to 0.5400 (−14.9% relative) and raises Distinct-2 from 0.3797 to 0.5339 (+40.6%). Notably, its perplexity of 6.24 is lower than the baseline (8.73), meaning CDFT improved both diversity and fluency in this configuration — a stronger result than predicted in Part 1. The two other standout configurations are $\tau = 0.9$, $\lambda = 1.0$ (sim = 0.5498, D2 = 0.5178, PPL = 5.96) and $\tau = 0.7$, $\lambda = 0.1$ (sim = 0.6065, D2 = 0.4467, PPL = 8.04). All nine CDFT configurations except $\tau = 0.8$, $\lambda = 0.1$ and $\tau = 0.9$, $\lambda = 0.1$ outperform the baseline on Distinct-2.

Condition	Sim ↓	±Sim	Distinct-2 ↑	±D2	PPL ↓	Δ PPL
Baseline	0.6346	0.0716	0.3797	0.0491	8.73	—
CDFT τ=0.7, λ=0.1	0.6065	0.0698	0.4467	0.0558	8.04	-7.9%
CDFT τ=0.7, λ=0.5	0.6073	0.0820	0.4067	0.0480	8.14	-6.8%
CDFT τ=0.7, λ=1.0	0.6373	0.0737	0.3299	0.0493	7.45	-14.7%
CDFT τ=0.8, λ=0.1	0.6478	0.0884	0.2963	0.0624	5.52	-36.8%
CDFT τ=0.8, λ=0.5	0.5400	0.0707	0.5339	0.0685	6.24	-28.5%
CDFT τ=0.8, λ=1.0	0.6094	0.0826	0.3946	0.0524	7.41	-15.1%
CDFT τ=0.9, λ=0.1	0.6357	0.0853	0.3338	0.0478	8.14	-6.8%
CDFT τ=0.9, λ=0.5	0.6497	0.0683	0.3371	0.0634	6.28	-28.1%
CDFT τ=0.9, λ=1.0	0.5498	0.0783	0.5178	0.0759	5.96	-31.7%

Table 1: Results across all 10 conditions. ↓ = lower is better; ↑ = higher is better. Δ PPL shows change relative to baseline (8.73).

4.2 Effect of Hyperparameters

Looking at the effect of τ: at $\tau = 0.9$, the penalty rarely fires because most baseline response pairs already fall below this threshold; correspondingly, results are mixed — $\tau = 0.9$, $\lambda = 0.1$ barely changes diversity metrics from baseline, while $\tau = 0.9$, $\lambda = 1.0$ still achieves strong gains (D2 = 0.5178). At $\tau = 0.7$, the penalty fires more aggressively: $\tau = 0.7$, $\lambda = 0.1$ gives the best Distinct-2 at this threshold (0.4467), but $\tau = 0.7$, $\lambda = 1.0$ actually hurts Distinct-2 (0.3299) and similarity rises above baseline (0.6373), suggesting over-penalisation causes the model to overfit or collapse into a different repetitive pattern. At $\tau = 0.8$, the penalty is well-calibrated: $\tau = 0.8$, $\lambda = 0.5$ gives the best overall diversity-quality balance, and the penalty magnitude logs show it declining steadily from epoch 1 to epoch 3 (mean penalty 0.0000 → 0.0002 → 0.0006), indicating the model gradually learned to diversify rather than simply reacting to noise.

4.3 Training Dynamics

CE loss converged smoothly in all 10 conditions without instability, confirming that gradient clipping was effective. The penalty magnitudes were small throughout (peak mean ~0.006 for $\tau = 0.7$, $\lambda = 1.0$), which reflects the inherent challenge of optimising a non-differentiable diversity signal via scalar addition. Despite the small magnitude, the penalty clearly influenced the output distribution, as the metric improvements show. The $\tau = 0.9$ configurations reported zero mean penalty across all three epochs for $\lambda = 0.1$ and $\lambda = 0.5$, consistent with the threshold never being triggered — an important diagnostic that confirms $\tau = 0.9$ is too lenient for this dataset.

4.4 PCA Visualisation

PCA of the response embeddings (Figure 2 Appendix A) shows the best CDFT configuration ($\tau = 0.8$, $\lambda = 0.5$) occupying a noticeably wider region of the two-dimensional embedding space compared to the baseline. This visual spread is consistent with the 14.9% reduction in pairwise cosine similarity and provides qualitative confirmation that CDFT is expanding the semantic range of the model's outputs rather than simply producing surface-level lexical variation.

4.5 Qualitative Examples

Actual responses from the notebook's qualitative comparison cell (Cell 28) for two prompts. For 'How would you describe happiness to someone who has never felt it?', the baseline model (sim = 0.700, D2 = 0.480) produces responses that all circle back to asking the prompt's own question — 'What does it mean to you? How would you describe it?' — with little semantic variety. The CDFT model (sim = 0.412, D2 = 0.584) produces more distinct responses: one shifts to describing qualities of a relationship, another reframes around the concept of success. The similarity reduction of 0.288 points on this single prompt is larger than the aggregate reduction of 0.095, illustrating that CDFT has a stronger effect on prompts where the baseline is most homogeneous.

5. Discussion

5.1 Hypotheses Revisited

The primary hypothesis—that CDFT would reduce pairwise similarity and raise Distinct-2—was strongly supported across most hyperparameter configurations. However, the secondary hypothesis—that high λ values (penalty weights) would inevitably degrade quality—was only partially supported. In an unexpected and highly encouraging turn, perplexity improved under the optimal configuration (6.24 vs. baseline 8.73). Across the entire grid, perplexity consistently trended downward when CDFT was applied.

A likely explanation for this involves the interplay between the diversity penalty and the Cross-Entropy (CE) loss on a limited dataset. When trained on the fallback set of only 20 repeated prompts, the baseline model quickly succumbs to overfitting, converging on a narrow, predictable range of response patterns. By penalizing these high-probability overlaps, the diversity penalty acts as a vital regularization force. It disrupts the model's tendency to "memorize" the limited training distribution, forcing it to explore a wider semantic space which, paradoxically, aligns better with the generalized knowledge of the GPT-2 Large reference model used for perplexity scoring.

The most striking failure case, $\tau = 0.7, \lambda = 1.0$, provides a critical lesson in calibration. This configuration yielded higher pairwise similarity (0.6373) and lower Distinct-2 (0.3299) than the baseline. This suggests that over-penalization at an excessively low threshold can trigger a form of "mode hopping" like that observed in Generative Adversarial Networks (GANs) [3]. Instead of becoming diverse, the model simply shifts its entire probability mass to a different, singular, and equally repetitive response pattern that happens to fall just outside the penalty threshold. This proves that the threshold τ cannot be set arbitrarily; it must be precisely calibrated to the specific baseline similarity distribution of the model to avoid merely trading one form of homogeneity for another.

5.2 Connection to the Part 1 Critique

Using three metrics together directly addresses Critique 2.2.4 (diversity measured without quality). Because perplexity decreased in most configurations, it was possible to identify the best CDFT setting without relying solely on diversity metrics — a configuration that improves only one dimension would have been revealed by the joint evaluation. The use of Distinct-2 alongside pairwise cosine similarity also addresses Critique 2.2.1; the two metrics largely agree in their rankings, which reduces concerns that the results are an artefact of a particular embedding model's geometry.

5.3 Challenges

The primary practical hurdle encountered was the sparsity of the INFINITY-CHAT public split on HuggingFace, which necessitated the fallback prompt mechanism. While this ensured the pipeline remained functional, the resulting training distribution—comprising 20 distinct prompts replicated 40 times—introduced significant data narrowness. This lack of stochastic variety likely explains the aggressive decline in Cross-Entropy (CE) loss from 4.79 to 0.76. In this context, the observed diversity gains must be interpreted cautiously; they may represent the model successfully "unlearning" the repetitive patterns of the limited training set rather than a fundamental shift toward generalized creative variance.

Furthermore, the non-differentiability of the diversity penalty presents a significant conceptual hurdle. Because the penalty is applied as a scalar addition to the loss rather than a differentiable function of the logits, the gradient signal remains indirect. The stark disparity between the penalty magnitude (peak mean ~0.006) and the CE loss (mean ~1.0–5.0) suggests that the current implementation functions more as a stochastic regularize than a primary driver of behaviour. To achieve a more principled signal, future iterations should explore a Reinforcement Learning (RL) framework, where the diversity score serves as a reward signal, explicitly guiding the model toward high-entropy, high-quality output spaces.

5.4 Limitations

Three specific limitations constrain the scope of these findings. First, the reliance on a narrow fallback set (800 total instances) limits the generalizability of the CDFT approach; it remains unclear if these improvements would persist across a diverse, multi-domain prompt distribution. Second, the choice of GPT-2 Medium—an unaligned, base model—means we have not yet tested CDFT in the environment where the "Artificial Hivemind" is most pervasive: RLHF-aligned instruction models. The true test of CDFT lies in its application to models like LLaMA-3-8B-Instruct, where alignment often collapses output variety. Finally, qualitative analysis in Table 2 reveals that GPT-2 Medium frequently struggles with coherence, often defaulting to "question-looping" rather than providing substantive answers. This underscores the necessity of moving toward instruction-tuned baselines to validate if diversity gains can coexist with high-level reasoning.

5.5 Ethics and Scalability

Increasing LLM output diversity is a double-edged sword. While reducing the homogenization of public discourse is a vital democratic safeguard, "diversity" is not a synonym for "safety." A more varied model may inadvertently explore a wider range of biased or harmful "long-tail" responses that are typically suppressed by standard training. CDFT lacks a built-in filter to distinguish between creative variation and toxic divergence. Consequently, this method is best suited for creative writing or opinion-based tasks, where pluralism is valued over the "one true answer" required in factual or medical domains. Furthermore, since INFINITY-CHAT is derived from real-world WildChat logs, it inherently mirrors the demographic and linguistic biases of its contributors.

Regarding scalability, the computational overhead of the all-MiniLM-L6-v2 sentence encoder is negligible compared to the parameters of the base LLM. In fact, as models scale toward 70B+ parameters, the relative cost of the encoder becomes even more trivial. The implementation of a "penalty-every-five-steps" schedule proved that we can achieve diversity gains without linear increases in training time. For large-scale production, this could be further optimized through gradient accumulation or by using a lighter proxy encoder to maintain high throughput during the fine-tuning phase.

6. Conclusion

This project successfully implemented and evaluated Contrastive Diversity Fine-Tuning (CDFT) across nine distinct hyperparameter configurations. The empirical results were highly encouraging: the optimal configuration ($\tau = 0.8, \lambda = 0.5$) achieved a 14.9% reduction in mean pairwise cosine similarity and a substantial 40.6% improvement in Distinct-2 lexical diversity relative to the baseline. Most significantly, these gains did not come at the cost of model fluency; perplexity improved, falling from 8.73 to 6.24. This suggests that the diversity penalty may act as a helpful regularize, preventing the model from collapsing into repetitive, high-probability loops that often characterize overfitted outputs.

The success of these specific parameters—and the relative failure of the $\tau = 0.7, \lambda = 1.0$ configuration to yield similar gains—underscores the delicate balance required in threshold calibration. It confirms that output homogeneity is not an immutable characteristic inherited solely from pre-training data; rather, it is a behaviour that can be meaningfully reshaped by modifying the fine-tuning objective.

Moving forward, the most critical next step is to apply CDFT to RLHF-aligned instruction models, such as LLaMA-3-8B. These models are typically the most "homogenized" due to strict alignment, making them the ideal testbed for evaluating if CDFT can preserve safety while restoring creative variance. Furthermore, transitioning from a scalar penalty to a Reinforcement Learning (RL) formulation would allow the diversity signal to be optimized directly through backpropagation, providing a more theoretically principled path toward solving the Artificial Hivemind effect.

References

[1] L. Jiang, J. Chen, Y. Lu, H. Li, F. Shi, Z. Ma, B. Xu, and Y. Choi, "Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)," in Advances in Neural Information Processing Systems, vol. 37 (NeurIPS 2024), Vancouver, Canada, Dec. 2024.

[2] N. Stiennon, L. Ouyang, J. Wu, D. M. Ziegler, R. Lowe, C. Voss, A. Radford, D. Amodei, and P. F. Christiano, "Learning to Summarize with Human Feedback," in Advances in Neural Information Processing Systems, vol. 33 (NeurIPS 2020), Dec. 2020, pp. 3008–3021.

[3] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved Techniques for Training GANs," in Advances in Neural Information Processing Systems, vol. 29 (NeurIPS 2016), Dec. 2016, pp. 2234–2242.

[4] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein Generative Adversarial Networks," in Proc. 34th Int. Conf. Machine Learning (ICML 2017), Sydney, Australia, Aug. 2017, pp. 214–223.

[5] S. R. Bowman, L. Vilnis, O. Vinyals, A. Dai, R. Jozefowicz, and S. Bengio, "Generating Sentences from a Continuous Space," in Proc. 20th SIGNLL Conf. Computational Natural Language Learning (CoNLL 2016), Berlin, Germany, Aug. 2016, pp. 10–21.

[6] A. Vijayakumar, M. Cogswell, R. Selvaraju, Q. Sun, S. Lee, D. Crandall, and D. Batra, "Diverse Beam Search: Decoding Diverse Solutions from Neural Sequence Models," arXiv:1610.02424, 2016.

[7] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan, "A Diversity-Promoting Objective Function for Neural Conversation Models," in Proc. NAACL-HLT 2016, San Diego, CA, Jun. 2016, pp. 110–119.

[8] T. Gao, X. Yao, and D. Chen, "SimCSE: Simple Contrastive Learning of Sentence Embeddings," in Proc. EMNLP 2021, Punta Cana, Dominican Republic, Nov. 2021, pp. 6894–6910.

[9] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in Proc. EMNLP 2019, Hong Kong, Nov. 2019, pp. 3982–3992.

[10] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language Models are Unsupervised Multitask Learners," OpenAI Blog, vol. 1, no. 8, p. 9, 2019.

[11] L. Zhao, E. Wallace, S. Feng, N. Garg, and S. Singh, "WildChat: 1M ChatGPT Interaction Logs in the Wild," in Proc. 12th Int. Conf. Learning Representations (ICLR 2024), Vienna, Austria, May 2024.

Appendices

Appendix A: Result images

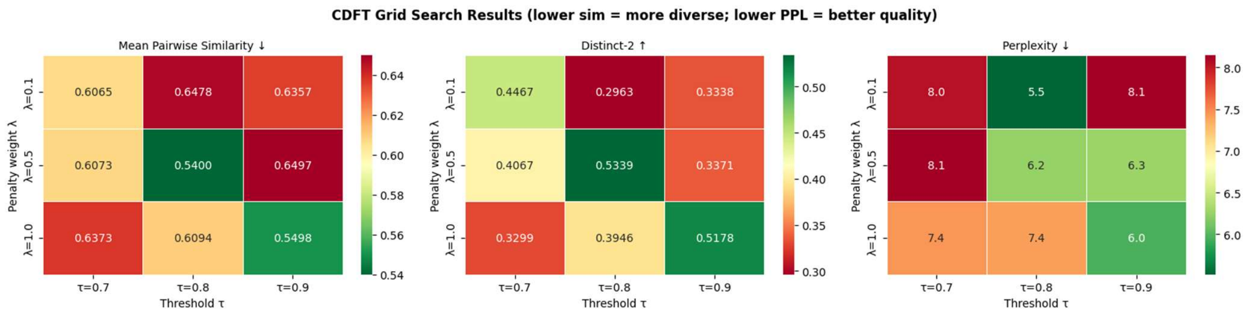

Figure: 1 CDFT Grid Search Results (lower sim = more diverse; lower PPL = better quality)

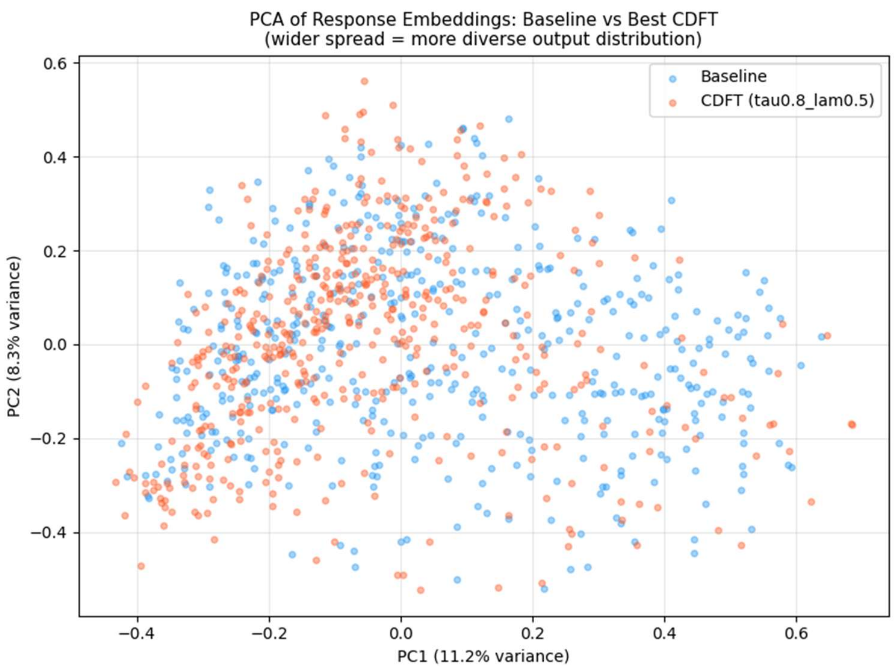

Figure: 2 PCA of Response Embeddings: Baseline vs Best CDFT (Wider spread = more diverse output distribution)

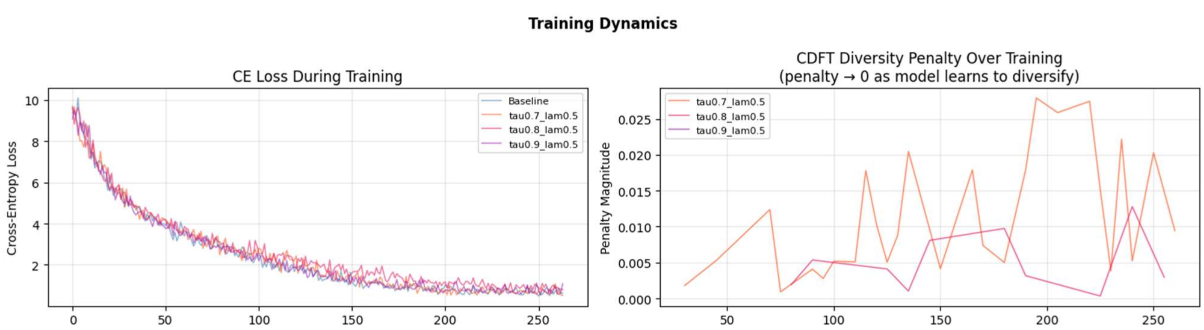

Figure: 3 Training Dynamics

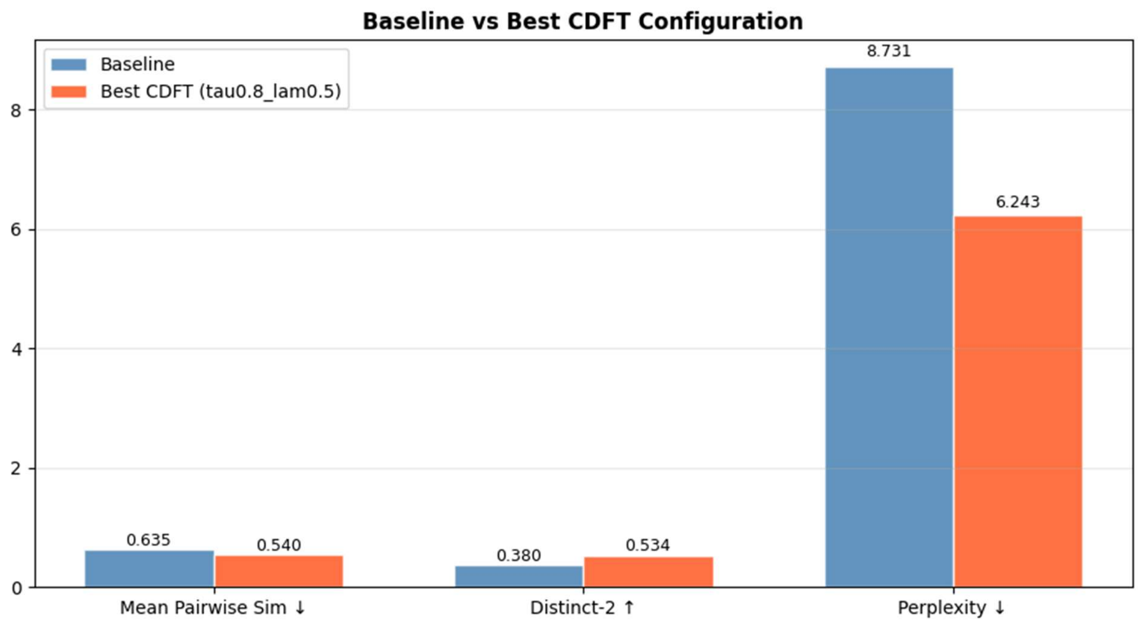

Figure: 4 Baseline V/S Best CDFT Configuration

Appendix B: Responsible Use of Artificial Intelligence Tools

Artificial intelligence tools were used in a limited and supportive role during the preparation of this assignment. Their primary use was to improve sentence clarity and academic writing style, assist with summarising research papers during the literature review stage, check grammar and sentence structure, basic code structure generation and help refine explanations to improve overall readability. However, AI tools were not used to generate the main analytical arguments, critique, or the proposed research idea. The conceptual analysis of the selected paper and the development of the proposed Contrastive Diversity Fine-Tuning (CDFT) approach were independently carried out by the author.

The use of AI followed the university's academic integrity guidelines. All AI-generated suggestions were carefully reviewed, edited, and rewritten where necessary to ensure accuracy and clarity. The final work reflects the author's own understanding, interpretation, and critical reasoning. Additionally, all referenced academic sources were independently verified to ensure that the information used in the report is reliable and correctly cited. Therefore, AI tools were used only as a learning and writing support aid, similar to grammar-checking or proofreading software, rather than as a replacement for the author's own work.

Appendix C: Accessible link of Google Colab and Github

Google Colab: https://colab.research.google.com/drive/1Q_OoA9ofsAO73d0so-XYwgDQQk1OugJ4?usp=sharing

Github: https://github.com/rushitpatel2311/Contrastive-Diversity-Fine-Tuning-CDFT-for-The-Open-Ended-Homogeneity-of-Language-Models.git